

A blurred background image of a business meeting. Several people in professional attire are gathered around a table. One person is holding a tablet displaying a document with charts and text. Another person is holding a smartphone. There are coffee cups on the table. The overall scene suggests a collaborative work environment.

Introduction to DML Statements

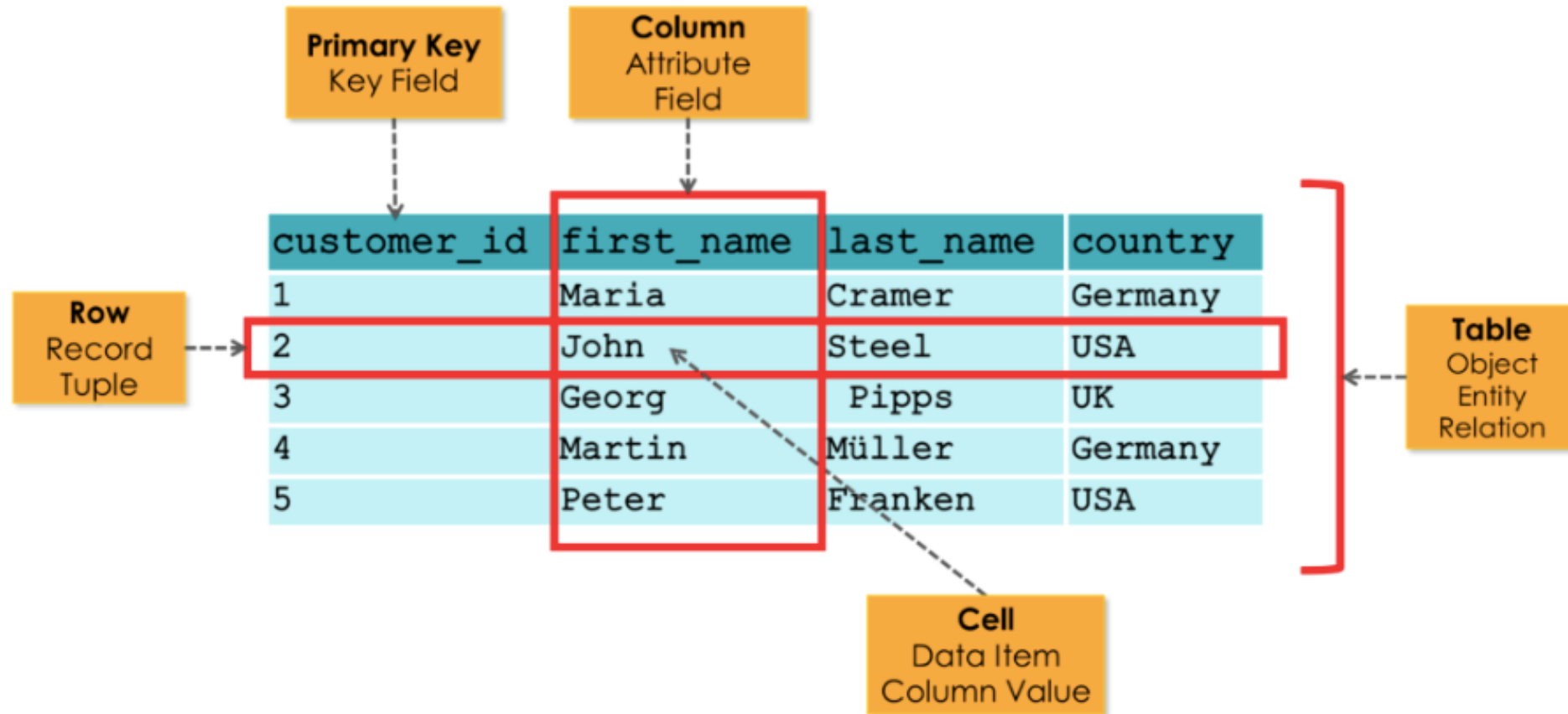
Objectives

- Discuss the different DML statements used.
- Discuss the different clauses used in SQL.
- Provide examples on how SQLs are used.

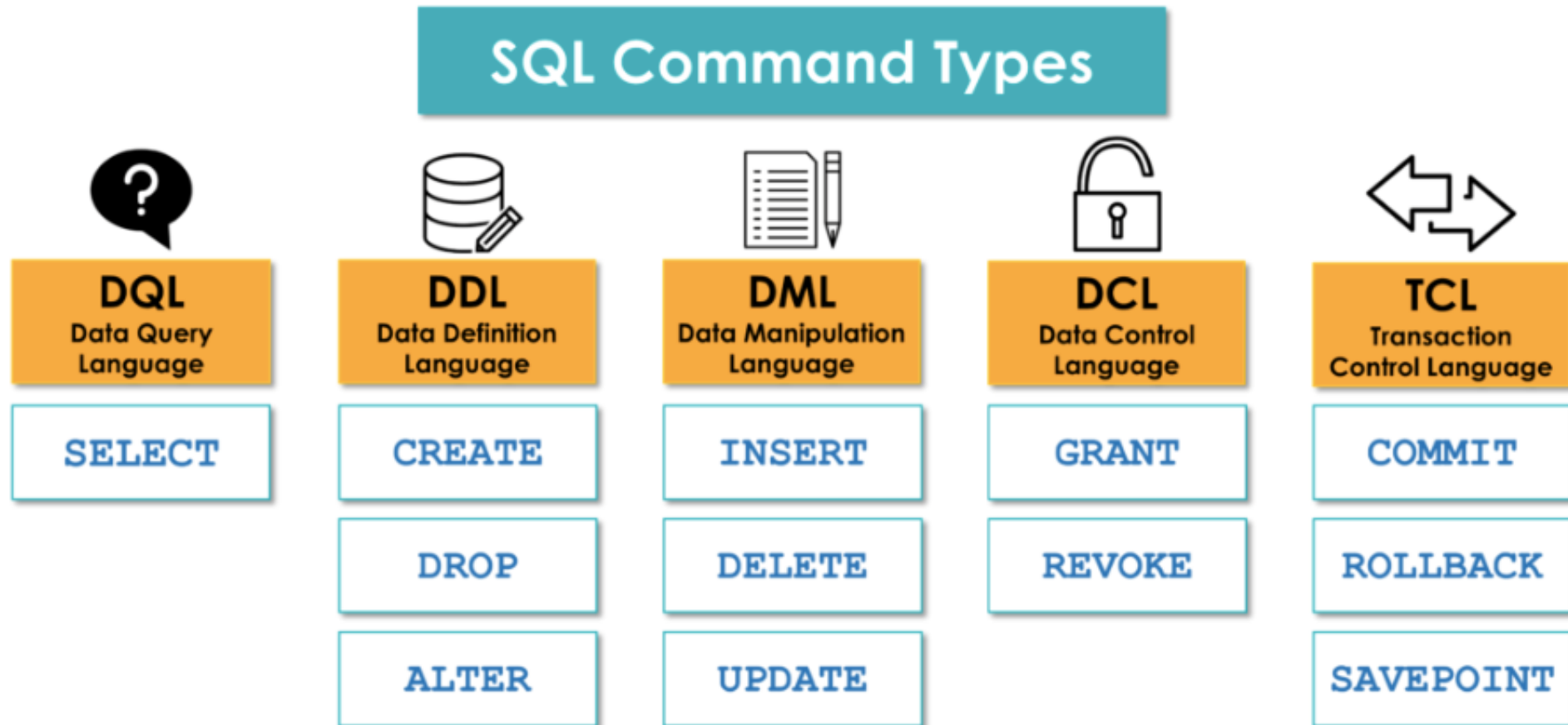
Objectives

- Discuss the different DML statements used.
- Discuss the different clauses used in SQL.
- Provide examples on how SQLs are used.

Review



SQL Commands



Data Query Language (DQL)

- **Data Query Language, (DQL)**, contains only one command called **SELECT**. It helps you to retrieve information from your database.
- **SELECT** is the most important command and concentrated focus of SQL. It contains many options and clauses which makes it very effective.

Data Manipulation Language (DML)

- **Data Manipulation Language, (DML)**, contains SQL commands used to manipulate your data inside your database.
- The **DML** command gives you tools to make changes to the records in a table, such as adding new records, deleting records, and updating records.
- The following commands considered to be the main DML commands:
 - **INSERT** — inserts data into a table.
 - **DELETE** — deletes data from a table.
 - **UPDATE** — updates existing data in a table.

ALIAS: Tables and Columns

Using the Schema name

```
105 • SELECT *  
106 FROM DBInventory.Products;  
107
```

100%	↺	15:103	
Result Grid   Filter Rows: Search			
	ProductID	Produ...	UnitPrice
▶	101	Laptop	500
▢	102	Mouse	20
	103	Keyb...	30
▢	104	Printer	100
	105	AVR	150
▢	105	AVR	150
	106	Mous...	5

What is an alias?

- SQL aliases are used to give a table, or a column in a table, a temporary name.
- Aliases are often used to make column names more readable.
- An alias only exists for the duration of that query.
- An alias is created with the AS keyword.

Note: In MySQL, you may or may not use “AS” when using aliases.

Example

```
SELECT ProductID as PI,  
       ProductName PN,  
       UnitPrice as UP  
from Products as P;
```

Result Grid   Filter Row				
	PI	PN	UP	
	101	Laptop	500	
▶	102	Mouse	20	
	103	Keyboard	30	
	104	Printer	100	
	105	AVR	150	
	105	AVR	150	
	106	Mousepad	5	

Use an ALIAS to:

1. Change a column or table name for improved readability within your code;
2. Change the output given to the recipients to address any formatting concerns.

SQL Clauses

What are Clauses in SQL?

- A clause in SQL is a built-in function that helps to fetch the required records from a database table.
- Clauses are in-built functions available to us in SQL.
- With the help of clauses, we can deal with data easily stored in the table.
- Clauses help us filter and analyze data quickly.
- When we have large amounts of data stored in the database, we use Clauses to query and get data required by the user.

Types of Clauses in SQL

Clause	Description
HAVING	HAVING clause can be used in a GROUP BY clause. It is used to specify a search condition for a group in the database tables.
WHERE	The WHERE clause in SQL is used to retrieve the specific data from the database that specifies the conditions exactly that are given in the UPDATE, DELETE, etc. statements.
ORDER BY	The ORDER BY clause in SQL is used for sorting the records of the database tables.
GROUP BY	To group the result set of the rows that have the same values in the result set from the database tables, the GROUP BY clause is used.
TOP	This clause is used when the database has many records. It is used to specify the total number of records to be fetched or returned.

Types of Clauses in SQL

WITH	WITH clause acts as a temporary view as it is available only during the execution of SELECT , UPDATE , INSERT , DELETE , etc. statements. It is used to simplify complex and long queries.
LIKE	The SQL LIKE clause compares a value to similar values using wildcard operators, i.e. per cent sign (%) and the underscore operator (_).
FROM	The FROM clause in SQL is used to select the database tables, which are manipulated using the SELECT , DELETE , and UPDATE statements.
LIMIT	The LIMIT clause is used when you are dealing with large databases. It is used to specify the maximum number of rows to be retrieved from the table.
AND	The AND clause is used when multiple conditions are specified in a query and returns a dataset when all the conditions given in the AND clause meet the requirements.
OR	The OR clause is used when multiple conditions are specified in a query and returns a dataset when one of those conditions gets satisfied.

The SQL WHERE Clause

- The WHERE clause is used to filter records.
- It is used to extract only those records that fulfill a specified condition.
- The WHERE clause in SQL is used to fetch the data or certain records that match the specified condition in the SELECT statement.
- SQL's WHERE clause is also used with the DELETE, UPDATE and other statements.

The WHERE Clause

Syntax:

- `SELECT column1, column2, ...
FROM table_name
WHERE condition;`

Example

Select all students in Computer Science in Department table

- `SELECT * FROM Department
WHERE Course='Computer Science';`

Operators in The WHERE Clause

- You can use operators other than the = operator to filter the search.
- Example
- Select all students with a StudentID greater than 4001:
- `SELECT * FROM Students
WHERE StudentID > 4001;`

The following operators can be used in the WHERE clause:

Operator	Description
=	Equal
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
<>	Not equal. Note: In some versions of SQL this operator may be written as !=
BETWEEN	Between a certain range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

Using BETWEEN

- The BETWEEN operator selects values within a given range.
- The values can be numbers, text, or dates.
- The BETWEEN operator is inclusive: begin and end values are included.

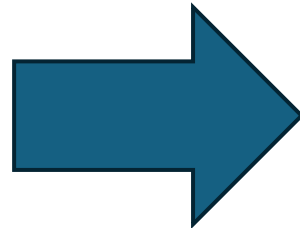
```
125 • SELECT * FROM Products
126     WHERE ProductID BETWEEN 102 AND 104;
127
```

100%	↕	37:126	
Result Grid   Filter Rows: Search			
	ProductID	Produ...	UnitPrice
	102	Mouse	20
	103	Keyb...	30
	104	Printer	100

Using BETWEEN

```
94 • SELECT *  
95 FROM Products;  
96
```

100%	↕	18:91
Result Grid		
Filter Row		
ProductID	Produ...	UnitPrice
101	Laptop	500
102	Mouse	20
103	Keyb...	30
104	Printer	100
105	AVR	150
105	AVR	150
106	Mous...	5



```
128 • SELECT * FROM Products  
129 WHERE ProductName BETWEEN "AVR" AND "Laptop";  
130
```

100%	↕	1:132
Result Grid		
Filter Rows: Search		
Export:		
ProductID	Produ...	UnitPrice
101	Laptop	500
103	Keyb...	30
105	AVR	150
105	AVR	150

Examples: Write the equivalent code

- Retrieve products with a specific ProductID greater than 102.
- Retrieve products where the UnitPrice is greater than 20.
- Retrieve products with product name as Laptop.
- Retrieve products with unit Price =50.
- Retrieve products where ProductID is less than 104.

Table: Products (Stores product-related information)

<u>ProductID</u>	ProductName	<u>UnitPrice</u>
101	Laptop	500
102	Mouse	20
103	Keyboard	30

ORDER BY

- The ORDER BY keyword is used to sort the result-set in ascending or descending order.
- To sort the database records, you can use the ORDER BY clause in SQL.
- This clause in SQL is used to arrange the fetched data in ascending or descending order based on the requirements.

ORDER BY

Syntax

- `SELECT column1, column2, ...
FROM table_name
ORDER BY column1, column2, ... ASC | DESC;`
- Order by clause is useful to get data in required sorting orders.
- By default, the data is sorted in ascending order.

ORDER BY

- Example

```
Select *  
from Employee  
Order by salary desc ;
```

The SQL AND Clause

- The WHERE clause can contain one or many AND operators.
- The AND operator is used to filter records based on more than one condition, like if you want to return all students from Computer Science that starts with the letter 'A'.

AND Syntax

- `SELECT column1, column2, ...
FROM table name
WHERE condition1 AND condition2 AND condition3
...;`
- if you want to return all students from Computer Science that starts with the letter 'A':
- `SELECT *
FROM Students
WHERE Course = 'Computer
Science' AND StudentName LIKE 'A%';`

AND Example

- `SELECT * FROM Customers
WHERE Country = 'Philippines'
AND City = 'Tagum'
AND PostalCode = 8100;`

AND Example

```
94 • SELECT *
95 FROM Products;
96
```

100% 18:91

Result Grid Filter Row

	ProductID	Produ...	UnitPrice
	101	Laptop	500
	102	Mouse	20
	103	Keyb...	30
	104	Printer	100
	105	AVR	150
	105	AVR	150
	106	Mous...	5



```
--
84 • SELECT *
85 FROM Products
86 WHERE ProductID=101
87 AND ProductName= 'Mouse';
88
```

100% 15:81

Result Grid Filter Rows: Search

	ProductID	Produ...	UnitPrice

OR Clause

- Or clause is beneficial when we need to pass multiple conditions, and we need data that satisfies any one of those specified conditions.
- When using an Or clause, we need to mention at least two conditions(the result would be at least one of the specified conditions.)

OR Clause- Usage

- Used while transacting and querying a database.
- Used in the update and delete statements.
- Or clause returns a data point when any one of the specified conditions is met.

OR Clause-Syntax

- `SELECT column1, column2, ...`
`FROM table name`
`WHERE condition1 OR condition2 OR condition3`
`...;`
- `SELECT *`
`FROM Students`
`WHERE Course = 'Computer Science' OR Course`
`= 'Information Technology';`

OR Example

```
94 • SELECT *
95 FROM Products;
96
```

100% 18:91

Result Grid Filter Row

	ProductID	Produ...	UnitPrice
	101	Laptop	500
	102	Mouse	20
	103	Keyb...	30
	104	Printer	100
	105	AVR	150
	105	AVR	150
	106	Mous...	5



```
--
89 • SELECT *
90 FROM Products
91 WHERE ProductID=100
92 OR ProductName= 'Mouse';
93
```

100% 16:87

Result Grid Filter Rows: Search

	ProductID	Produ...	UnitPrice
▶	102	Mouse	20

AND vs OR

- The AND operator displays a record if *all* the conditions are TRUE.
- The OR operator displays a record if *any* of the conditions are TRUE.

AND vs OR Example

- Make sure you use parenthesis to get the correct result.
- Example:

Select all Filipino customers that starts with either "A" or "B":

- ```
SELECT * FROM Customers
WHERE Country
= 'Philippines' AND (CustomerName LIKE 'A%' OR CustomerName LIKE 'B%');
```

# AND vs OR Example

Note:

- Without parenthesis, the select statement will return all customers from Spain that starts with a "A", *plus* all customers that starts with an "B", regardless of the country value:
-

# Filter Multiple Values at once (IN)

- USE IN Keyword to:
  1. Filter a single column with multiple values
  2. Improve the readability of your code
  3. Make changes to your conditions more efficiently when required.

```
116 • SELECT * FROM Products
117 WHERE ProductID IN (101, 102, 103);
118
```

| 100%                                                                                                                                                                                                                                 | ↺         | 16:112   |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|-----------|
| Result Grid   Filter Rows: <input type="text" value="Search"/> |           |          |           |
|                                                                                                                                                                                                                                      | ProductID | Produ... | UnitPrice |
| ▶                                                                                                                                                                                                                                    | 101       | Laptop   | 500       |
|                                                                                                                                                                                                                                      | 102       | Mouse    | 20        |
|                                                                                                                                                                                                                                      | 103       | Keyb...  | 30        |

# SQL NOT Operator

- The NOT operator is used in combination with other operators to give the opposite result, also called the negative result.

# SQL NOT Operator

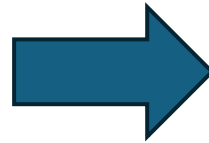
- Syntax
- `SELECT column1, column2, ...  
FROM table_name  
WHERE NOT condition;`
- Select only the students that are NOT from Computer Science:

```
SELECT * FROM Students
WHERE NOT Course = 'Computer Science';
```

# SQL NOT Operator

```
119 • SELECT * FROM Products
120 WHERE ProductName="Laptop";
121
```

| 100%                | ↺        | 18:116    |
|---------------------|----------|-----------|
| Result Grid         |          |           |
| Filter Rows: Search |          |           |
| ProductID           | Produ... | UnitPrice |
| ▶ 101               | Laptop   | 500       |



```
119 • SELECT * FROM Products
120 WHERE NOT ProductName="Laptop";
121
```

| 100%                | ↺        | 11:116    |
|---------------------|----------|-----------|
| Result Grid         |          |           |
| Filter Rows: Search |          |           |
| ProductID           | Produ... | UnitPrice |
| 102                 | Mouse    | 20        |
| 103                 | Keyb...  | 30        |
| 104                 | Printer  | 100       |
| 105                 | AVR      | 150       |
| 105                 | AVR      | 150       |
| 106                 | Mous...  | 5         |

# SQL NOT Operator and <> Operator

```
122 • SELECT * FROM Products
123 WHERE NOT ProductID IN (101, 102, 103);
124
```

| 100%                                                                                                                                                                                                                                                 | ↕         | 23:119   |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|-----------|
| Result Grid   Filter Rows: <input type="text" value="Search"/> <span>Export</span> |           |          |           |
|                                                                                                                                                                                                                                                      | ProductID | Produ... | UnitPrice |
|                                                                                                                                                                                                                                                      | 104       | Printer  | 100       |
|                                                                                                                                                                                                                                                      | 105       | AVR      | 150       |
| ▶                                                                                                                                                                                                                                                    | 105       | AVR      | 150       |
|                                                                                                                                                                                                                                                      | 106       | Mous...  | 5         |

```
131 • SELECT * FROM Products
132 WHERE ProductID <> 104;
133
```

```
131 • SELECT * FROM Products
132 WHERE ProductID != 104;
133
```

| 100%                                                                                                                                                                                                                                 | ↕         | 24:132   |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|-----------|
| Result Grid   Filter Rows: <input type="text" value="Search"/> |           |          |           |
|                                                                                                                                                                                                                                      | ProductID | Produ... | UnitPrice |
|                                                                                                                                                                                                                                      | 101       | Laptop   | 500       |
|                                                                                                                                                                                                                                      | 102       | Mouse    | 20        |
|                                                                                                                                                                                                                                      | 103       | Keyb...  | 30        |
|                                                                                                                                                                                                                                      | 105       | AVR      | 150       |
|                                                                                                                                                                                                                                      | 105       | AVR      | 150       |
|                                                                                                                                                                                                                                      | 106       | Mous...  | 5         |

# What is a NULL Value?

- A field with a NULL value is a field with no value.
- If a field in a table is optional, it is possible to insert a new record or update a record without adding a value to this field. Then, the field will be saved with a NULL value.

# What is a NULL Value?

- Syntax:
- `SELECT column_names  
FROM table_name  
WHERE column_name IS NULL;`

# What is a NULL Value?

```
94 • SELECT *
95 FROM Products;
96
```

|             |           |          |           |
|-------------|-----------|----------|-----------|
| 100%        | ↕         | 18:91    |           |
| Result Grid |           |          |           |
|             |           |          |           |
|             | ProductID | Produ... | UnitPrice |
|             | 101       | Laptop   | 500       |
|             | 102       | Mouse    | 20        |
|             | 103       | Keyb...  | 30        |
|             | 104       | Printer  | 100       |
|             | 105       | AVR      | 150       |
|             | 105       | AVR      | 150       |
|             | 106       | Mous...  | 5         |



```
134 • SELECT * FROM Products
135 WHERE ProductID IS NULL;
136
```

|             |           |          |           |
|-------------|-----------|----------|-----------|
| 100%        | ↕         | 1:130    |           |
| Result Grid |           |          |           |
|             |           |          |           |
|             | ProductID | Produ... | UnitPrice |



```
134 • SELECT * FROM Products
135 WHERE ProductID IS NOT NULL;
136
```

|             |           |          |           |
|-------------|-----------|----------|-----------|
| 100%        | ↕         | 24:132   |           |
| Result Grid |           |          |           |
|             |           |          |           |
|             | ProductID | Produ... | UnitPrice |
| ▶           | 101       | Laptop   | 500       |
|             | 102       | Mouse    | 20        |
|             | 103       | Keyb...  | 30        |
|             | 104       | Printer  | 100       |
|             | 105       | AVR      | 150       |
|             | 105       | AVR      | 150       |
|             | 106       | Mous...  | 5         |

# Filtering Values Using LIKE

- The LIKE operator is used in a WHERE clause to search for a specified pattern in a column.
- There are two wildcards often used in conjunction with the LIKE operator:
  - The percent sign % represents zero, one, or multiple characters
  - The underscore sign \_ represents one, single character

# Filtering Values Using LIKE

## *The \_ Wildcard*

- The \_ wildcard represents a single character.
- It can be any character or number, but each \_ represents one, and only one, character.

```
137 • SELECT * FROM Products
138 WHERE ProductName LIKE 'L_p__p';
139
```

|             |           |                                                                                                                                                                                      |                                     |
|-------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 100%        | ↕         | 33:138                                                                                                                                                                               |                                     |
|             |           |                                                                                                                                                                                      |                                     |
| Result Grid |           |   Filter Rows: | <input type="text" value="Search"/> |
|             | ProductID | Produ...                                                                                                                                                                             | UnitPrice                           |
|             | 101       | Laptop                                                                                                                                                                               | 500                                 |

# Filtering Values Using LIKE

## The % Wildcard

- The % wildcard represents any number of characters, even zero characters.

```
140 • SELECT * FROM Products
141 WHERE ProductName LIKE 'M%';
142
```

100% 27:141

**Result Grid**   Filter Rows:

|  | ProductID | Produ... | UnitPrice |  |
|--|-----------|----------|-----------|--|
|  | 102       | Mouse    | 20        |  |
|  | 106       | Mous...  | 5         |  |

# Filtering Values Using LIKE

```
140 • SELECT * FROM Products
141 WHERE ProductID LIKE '10%';
142
```

| 100%                                                                                                                                                                                                                             | ↕         | 22:137   |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|-----------|
| Result Grid   Filter Rows: <input type="text" value="Search"/> |           |          |           |
|                                                                                                                                                                                                                                  | ProductID | Produ... | UnitPrice |
| ▶                                                                                                                                                                                                                                | 101       | Laptop   | 500       |
| ▢                                                                                                                                                                                                                                | 102       | Mouse    | 20        |
|                                                                                                                                                                                                                                  | 103       | Keyb...  | 30        |
| ▢                                                                                                                                                                                                                                | 104       | Printer  | 100       |
|                                                                                                                                                                                                                                  | 105       | AVR      | 150       |
| ▢                                                                                                                                                                                                                                | 105       | AVR      | 150       |
|                                                                                                                                                                                                                                  | 106       | Mous...  | 5         |